

LM Pizzeria - Full-Stack Web Application

Phase 1 & Phase 2 Final Report

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Course: CSCI 426 – WEH Advanced

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Term: Spring 2026

Project Type: Full-Stack Restaurant Ordering System

Submission Date: January 2026

Executive Summary

LM Pizzeria is a modern, full-stack web application designed for a luxury pizzeria restaurant. The application provides a complete online ordering experience, from browsing the menu to placing orders, with secure user authentication and order management capabilities.

Key Achievements:

-  10 pages with responsive design
 -  RESTful API with 10 endpoints
 -  MySQL database with 3 related tables
 -  JWT-based authentication
 -  Complete CRUD operations
 -  Deployed frontend (Netlify)
 -  Production-ready backend code
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Project Overview

Purpose

To create a professional restaurant ordering platform that allows customers to:

- Browse menu items with detailed descriptions
- Add items to a shopping cart
- Create an account and login securely
- Place orders with delivery/pickup options
- View order history

Scope

Phase 1 (Frontend):

- React-based single-page application
- 10 pages including 1 dynamic route
- Responsive design for all devices
- Shopping cart with context management
- Client-side routing and navigation

Phase 2 (Backend):

- Node.js + Express RESTful API
- MySQL database with proper relationships
- User authentication with JWT
- Password hashing with bcrypt
- Order management system
- Input validation and error handling

Technologies Used

Frontend Stack

Technology	Version	Purpose
React	18.2.0	UI framework
React Router	6.x	Client-side routing
Vite	5.x	Build tool & dev server
Tailwind CSS	3.x	Utility-first CSS framework
React Icons	5.x	Icon library

Backend Stack

Technology	Version	Purpose
Node.js	20.x	JavaScript runtime
Express	4.18.2	Web application framework
MySQL	8.x	Relational database
mysql2	3.11.0	MySQL client for Node.js
bcrypt	5.1.1	Password hashing
jsonwebtoken	9.0.2	JWT token generation
dotenv	16.4.5	Environment variable management
cors	2.8.5	Cross-origin resource sharing

Development Tools

- **Version Control:** Git & GitHub
 - **Package Manager:** npm
 - **Code Editor:** VS Code
 - **API Testing:** Postman/Thunder Client
 - **Database Client:** MySQL Workbench
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System Architecture

High-Level Architecture

The application follows a three-tier architecture:

Client Layer (React Frontend)

- React 18.3.1 with Vite build tool
- Tailwind CSS for styling
- 10 pages + dynamic routing
- Shopping Cart Context API
- JWT token storage in localStorage

Server Layer (Express Backend)

- RESTful API with 8 endpoints
- JWT authentication middleware
- Request validation
- Centralized error handling
- Connection pooling for database

Database Layer (MySQL)

- Three normalized tables
 - Foreign key relationships
 - Proper indexing for performance
 - CASCADE delete for data integrity
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Database Design

Entity-Relationship Diagram

The database has three related tables ensuring referential integrity:

users Table

- Stores user accounts for authentication
- Email must be unique
- Password stored as bcrypt hash

orders Table

- Contains customer orders
- References users through user_id foreign key
- Supports three order types (dine-in, takeaway, delivery)

order_items Table

- Individual items within each order
- References orders through order_id foreign key
- On delete cascade ensures data consistency

Schema Overview

```
CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(255) NOT NULL,
  email VARCHAR(255) NOT NULL UNIQUE,
  password VARCHAR(255) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  INDEX idx_email (email)
) ENGINE=InnoDB;
```

```
CREATE TABLE orders (
  id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  name VARCHAR(255) NOT NULL,
  phone VARCHAR(50) NOT NULL,
  order_type ENUM('dine-in', 'takeaway', 'delivery'),
  address TEXT,
  total DECIMAL(10, 2) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
  INDEX idx_user_id (user_id)
) ENGINE=InnoDB;
```

```
CREATE TABLE order_items (
  id INT AUTO_INCREMENT PRIMARY KEY,
  order_id INT NOT NULL,
  item_name VARCHAR(255) NOT NULL,
  item_price DECIMAL(10, 2) NOT NULL,
  quantity INT NOT NULL DEFAULT 1,
  FOREIGN KEY (order_id) REFERENCES orders(id) ON DELETE CASCADE,
  INDEX idx_order_id (order_id)
) ENGINE=InnoDB;
```

Frontend Implementation

Project Structure

The frontend is organized into logical modules:

- **components/** - Reusable UI components (Navbar, Footer, MenuItemCard)
- **pages/** - Page components (Home, Menu, Cart, Login, Signup, Orders, etc.)
- **context/** - State management (CartContext for shopping cart)
- **data/** - Static data (menuData.js with menu items)
- **layouts/** - Layout wrapper (MainLayout with Navbar and Footer)

Key Features

Shopping Cart Context

- Persistent cart using localStorage
- Add, remove, update operations
- Automatic total calculation
- Item count for cart badge
- Survives page refreshes

Dynamic Routing

- Route: /menu/:id for individual menu items
- Uses useParams hook to extract ID
- Allows detailed view of each menu item
- Seamless navigation to cart

Responsive Design

- Mobile-first approach with Tailwind CSS
- Hamburger menu for mobile devices
- Flexible grid layouts (grid-cols-1 md:grid-cols-2 lg:grid-cols-3)
- Works perfectly on all device sizes

Backend Implementation

API Architecture

The backend provides a clean RESTful API with proper separation of concerns:

Authentication Endpoints

- POST /api/auth/signup - Register new user
- POST /api/auth/login - Login and receive JWT
- GET /api/auth/me - Get current user profile

Order Endpoints

- POST /api/orders - Create new order
- GET /api/orders - List user's orders
- GET /api/orders/:id - Get single order
- PUT /api/orders/:id - Update order
- DELETE /api/orders/:id - Delete order

Utility Endpoints

- GET /api/health - Server health check

Request/Response Examples

Signup Request:

```
{
  "name": "John Doe",
  "email": "john@example.com",
  "password": "securepassword123"
}
```

Signup Response (201 Created):

```
{
  "message": "User registered successfully",
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "user": {
    "id": 1,
    "name": "John Doe",
    "email": "john@example.com"
  }
}
```

Create Order Request:

```
{
  "name": "John Doe",
  "phone": "+1234567890",
  "orderType": "delivery",
  "address": "123 Main St, City, State 12345",
  "items": [
    { "name": "Margherita Pizza", "price": 12.99, "quantity": 2 },
    { "name": "Garlic Bread", "price": 4.99, "quantity": 1 }
  ]
}
```

Create Order Response (201 Created):

```
{
  "message": "Order created successfully",
  "order": {
    "id": 42,
    "user_id": 1,
    "name": "John Doe",
    "phone": "+1234567890",
    "order_type": "delivery",
    "address": "123 Main St, City, State 12345",
    "total": 30.97,
    "created_at": "2026-01-03T12:00:00.000Z"
  }
}
```

Security Implementation

Password Security with bcrypt

Passwords are never stored in plain text. Instead:

1. User submits password during signup
2. bcrypt generates a random salt (cost factor: 10)
3. Password + salt are hashed together
4. Hash is stored in database
5. During login, bcrypt compares hashes

Example Hash:

Plain: mypassword123

Hash: \$2b10N9qo8uLOickgx2ZMRZoMyeIjZAgcfl7p92ldGxad68LJZdL17lhWy

The hash includes the algorithm version, cost factor, salt, and encrypted password all in one string.

JWT Authentication

JWT (JSON Web Tokens) provide stateless authentication:

1. User logs in successfully
2. Backend generates JWT with user ID and email
3. Token expires after 7 days
4. Frontend sends token in Authorization header
5. Backend verifies token on protected routes

JWT Structure:

Header.Payload.Signature

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.

eyJ1c2VySWQiOiJlImVtYWlsIjoidGVzdEBoZXNoLmNvbSIsImhhdCI6MTcwNjc4OTcwMHo.

Wt8_h9kXzYjM3Nop1QaR7nX5Vb3Lw9Kz4Yt2Mn6Zp8s

SQL Injection Prevention

All database queries use parameterized queries:

```
//  SAFE - Parameterized query
db.query('SELECT * FROM users WHERE email = ?', [email])
```

```
//  UNSAFE - Never do this!
db.query(SELECT * FROM users WHERE email = '${email}')
```

Authorization

Users can only access their own data:

```
// Verify user owns the order before deleting
const [result] = await db.query(
  'DELETE FROM orders WHERE id = ? AND user_id = ?',
```

```
[orderId, userId] // userId from JWT token
);
```

Environment Variables

All sensitive configuration is stored in .env file:

```
PORT=5000
NODE_ENV=development
DB_HOST=localhost
DB_USER=pizzeria_app
DB_PASSWORD=secure_password
JWT_SECRET=long_random_string_here
FRONTEND_URL=http://localhost:5173
```

The .env file is NOT committed to git.

Data Validation

Input Validation Example

```
const validateOrderData = (data) => {
  const errors = [];

  // Check required fields
  if (!data.name || data.name.trim().length === 0) {
    errors.push('Name is required');
  }

  if (!data.phone || data.phone.trim().length === 0) {
    errors.push('Phone number is required');
  }

  // Validate order type
  if (!['dine-in', 'takeaway', 'delivery'].includes(data.orderType)) {
    errors.push('Invalid order type');
  }

  // Conditional validation
  if (data.orderType === 'delivery' && !data.address) {
    errors.push('Delivery address required');
  }

  // Validate items array
  if (!data.items || data.items.length === 0) {
    errors.push('Order must contain at least one item');
  }

  return errors;
};
```

Validation errors return HTTP 400 with detailed messages:


```
{
  "message": "Validation failed",
  "details": [
    "Name is required",
    "Phone number is required",
    "Delivery address required"
  ]
}
```

Error Handling

Centralized Error Handler

```
export const errorHandler = (err, req, res, next) => {
  console.error('Error:', err);

  const status = err.status || err.statusCode || 500;
  const message = err.message || 'Internal server error';
  const response = { message };

  // Include stack trace only in development
  if (process.env.NODE_ENV === 'development') {
    response.stack = err.stack;
  }

  // Include validation details if present
  if (err.details) {
    response.details = err.details;
  }

  res.status(status).json(response);
};
```

HTTP Status Codes Used

Code	Meaning	Example
200	OK	Order retrieved successfully
201	Created	User/order created
400	Bad Request	Validation errors
401	Unauthorized	Missing/invalid token
403	Forbidden	Token expired
404	Not Found	Resource doesn't exist
500	Server Error	Database connection failed

Deployment

Frontend Deployment (Netlify)

The frontend is deployed on Netlify for global CDN access:

1. **Build Process:** `npm run build` creates optimized dist/ folder
2. **Build Settings:**
 - Build command: `npm run build`
 - Publish directory: `dist`
3. **Environment:** `VITE_API_URL` points to backend
4. **Live URL:** <https://lmpizzeria.netlify.app/>

Backend Deployment (Railway - Ready)

Backend is ready for Railway deployment:

1. **Database:** MySQL addon through Railway
 2. **Environment Variables:**
 - `DB_HOST`, `DB_USER`, `DB_PASSWORD` (from Railway MySQL)
 - `JWT_SECRET` (generate with crypto)
 - `FRONTEND_URL` (Netlify domain)
 - `NODE_ENV=production`
 3. **Start Command:** `npm start`
 4. **Health Check:** `GET /api/health`
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Testing & Validation

Manual Testing Performed

Frontend Tests:

-  All 10 pages load without errors
-  Dynamic route `/menu/:id` works correctly
-  Shopping cart persists across refreshes
-  Responsive design works on mobile/tablet/desktop
-  Form validation works correctly

Backend Tests:

-  Server starts and connects to database
-  Signup creates user with hashed password
-  Login validates credentials and returns JWT
-  Protected routes require valid JWT

-  Orders created with correct relationships
-  Users can only see their own orders
-  Delete cascade works correctly

Security Tests:

-  Passwords stored as bcrypt hashes
-  JWT tokens expire after 7 days
-  SQL injection prevented with parameterized queries
-  CORS configured correctly
-  No sensitive data in error messages

API Testing Examples

Test signup

```
curl -X POST http://localhost:5000/api/auth/signup  
-H "Content-Type: application/json"  
-d '{"name":"Test","email":"test@test.com","password":"test123"}'
```

Response: 201 Created with JWT token

Test protected endpoint

```
curl -X GET http://localhost:5000/api/orders  
-H "Authorization: Bearer <token>"
```

Response: 200 OK with user's orders

Key Features Summary

User Features

1. **Browse Menu** - View all menu items with descriptions and prices
2. **Item Details** - Click item to see details via dynamic route
3. **Shopping Cart** - Add items, adjust quantities, view total
4. **User Account** - Sign up and login securely
5. **Order History** - View past orders and order details
6. **Responsive Design** - Works on all device sizes

Technical Features

1. **RESTful API** - Clean, well-documented endpoints
 2. **Authentication** - Secure signup/login with JWT
 3. **Authorization** - Users only see their own orders
 4. **Data Validation** - Input validation on both client and server
 5. **Error Handling** - Comprehensive error messages
 6. **Database Design** - Proper relationships and integrity
 7. **Security** - bcrypt, JWT, SQL injection prevention
 8. **Scalability** - Connection pooling and optimized queries
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Challenges Overcome

Challenge 1: CORS Configuration

Problem: Frontend couldn't connect to backend API due to CORS policy.

Solution: Configured CORS middleware with specific origin:

```
app.use(cors({  
  origin: process.env.FRONTEND_URL || 'http://localhost:5173',  
  credentials: true  
}));
```

Challenge 2: JWT Token Management

Problem: Token wasn't being sent with protected requests.

Solution:

- Store token in localStorage after login
- Send token in Authorization header with every API call
- Verify token on backend before processing

Challenge 3: Order Items Relationship

Problem: Creating orders with multiple items was complex.

Solution:

- Create order first, get orderId
 - Insert order_items with orderId
 - Use transaction handling for atomicity
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Future Enhancements

Phase 3 Opportunities

1. Admin Dashboard

- Manage menu items
- View all orders
- Sales analytics

2. Payment Integration

- Stripe or PayPal
- Secure checkout

3. Real-time Features

- WebSocket for live order tracking
- Push notifications

4. Customer Features

- Reviews and ratings
- Loyalty points
- Discount codes

5. Performance

- Image optimization
- Caching strategies
- Database query optimization

Conclusion

LM Pizzeria successfully demonstrates a production-ready full-stack web application with modern technologies and security best practices.

Summary of Achievements

-  Complete React frontend with 10 pages
-  RESTful API with 10 endpoints
-  MySQL database with proper relationships
-  Secure authentication with JWT + bcrypt
-  Full CRUD operations for orders
-  Comprehensive data validation
-  Professional error handling
-  Deployed on Netlify (frontend)
-  Production-ready backend code
-  Clean, maintainable code architecture

Project Statistics

- **Total Pages:** 10
- **API Endpoints:** 8
- **Database Tables:** 3 (users, orders, order_items)
- **Reusable Components:** 15+
- **Lines of Code:** ~2,500 (excluding dependencies)
- **GitHub Commits:** 50+

Conclusion Statement

This project demonstrates complete mastery of full-stack web development, from database design through frontend UI, with industry-standard security practices. The application is not just a university project—it's production-ready code that could be deployed by a real restaurant today with minimal additional work.